

HW 8

Exercise 1 (Data set: `A.csv`).

Fit a first-order model for y as a function of x_1 , x_2 and x_3 .

Check the assumption of a constant error variance by plotting the model residuals against predicted arsenic level. Interpret the results.

Exercise 2 (Data set: `G.csv`).

Consider the interaction model $E(y) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1x_2$.

- a. Fit the model to the data, then plot the residuals against predicted y .
- b. Is the assumption of a constant error variance reasonably satisfied? If not, suggest how to modify the model.
- c. Use a residual graph to check the assumption of normal errors for the interaction model for y . Is the normality assumption reasonably satisfied? If not, suggest how to modify the model.
- d. Identify any outliers in the interaction model for y . Are any of these outliers influential data points? If so, what are your recommendations?